

Leyan Wu

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Education

Peking University, B.S. Candidate in Physics & Economics Expected Jun. 2028

- **Selected Coursework:** Econometrics (100/100), Topics in Quantitative Finance, Linear Algebra (98/100), Complex Analysis (98/100), Calculus (97/100), Data Structures and Algorithms (97/100), Computer Vision and Deep Learning, and Statistical Mechanics
- **GPA:** 3.914/4.00; **Ranked 1st of 28 in the major; top 1% in the college (all majors)**

Technical Skills

- **Programming:** Python (NumPy, pandas, SciPy, Matplotlib), Bash, Git
- **Statistics:** Linear regression, maximum likelihood, hypothesis testing, stochastic processes, portfolio optimization, and option pricing
- **Scientific Computing:** Signal processing, parameter estimation, curve fitting, and DFT with VASP

Research and Internship Experience

DP Technology AI for Science Intern Aug. 2026 – Present

- Building an agentic research harness with human-review gates, immutable checkpoints, automated validation, and error-specific rollback.
- Designed a pipeline that converts papers and coarse claim graphs into traceable, executable reasoning graphs.

Institute of Condensed Matter Physics, PKU Undergraduate Researcher Aug. 2025 – Present

- Designed and fabricated dual-gate devices for magnetotransport measurements in 2D heterostructures.
- Analyzed transport data using background subtraction, symmetry decomposition, and cross-device comparison across gate voltage, magnetic field, and layer parity to isolate interfacial-coupling effects.

Center for Fundamental Physics Experiments, PKU Feb. 2025 – Jul. 2025

- Modeled Rabi splitting and PT-symmetry breaking with coupled oscillators.
- Estimated parameters from noisy spectra via nonlinear least squares and validated the transition criterion against numerical simulations and 5–25 kHz experimental response data.

Honors and Leadership

- Peking University Yanchuang Group Scholarship (Top 10 in college)
- Peking University Freshman Scholarship, Second Prize
- Peking University Outstanding Student Award
- Director, Department of Academic and Innovation, Peking University Student Union